B.TECH COMPUTER ENGINEERING

8TH SEMESTER

MAJOR PROJECT SYNOPSIS

TITLE

Smart eyeglass with face recognition, obstacle detection features for visually Impaired

PROJECT SUPERVISOR

Prof. Mohammad Amjad

SUBMITTED BY

Md Minhaaz 19BCS014

Shahabuddin 19BCS031

SMART EYEGLASS

1. ABSTRACT:

Visually impaired and blind people require constant assistance from others in order to move around, making them extremely dependent on others in their social lives. For visually impaired and blind people, object detection and distance calculation are major challenges. Earlier navigation systems were both costly and time-consuming to use in everyday life. This study describes a blind or visually impaired person's electronic navigation system (subject). Using an ultrasonic sensor network, this system detects objects. It effectively measures the distance between the person and the detected obstacle(object). It employs speech feedback to inform the user of the location and distance of the indicated obstacle. Speaker is used to provide such voice messages to the subject. The project's objective is to equip visually impaired and blind people with a more cost-effective and portable navigation system.

2. INTRODUCTION:

Globally, there are 2.2 billion people who are visually impaired or blind, according to 2022 World Health Organization research on disability. As the population is growing older and as people's habits change, chronic eye diseases like diabetic retinopathy are likely to become more common.

Eyes are the most important organ to perceive the outside world. Without eyes it is difficult to move. People who are blind or visually impaired mostly rely on other people to move around.

Earlier it was difficult to create a sophisticated navigation system for such people because it was too costly and time consuming. Moreover, most of the sensors were of huge sizes, which made it difficult to use it on basis. But due to advancements in technology in recent times, now sensors have become cheaper, and their size has also been reduced.

It is now possible to create a eyeglass using ultrasonic sensors and ML, which will help visually impaired people to move around without being dependent on other people. As soon as an obstacle comes in way, the glasses will alert the user using a voice, so that user can know that there is an obstacle at some distance and user can avoid it. We'll also include facial recognition, so that if a known person comes in front of a person, that person can easily recognize another person.

3. PROPOSED METHOD:

For this project we'll use 2 different codes in Python3. One to tell the distance of obstacles ahead, and other for face recognition.

- **Code for Obstacle Detection:**

Open the Python3 IDE to start coding. Here we need two modules – time and espeak. We will import both the modules in code. After that we created a distance function where we import the GPIO and set the pins for distance sensor. Next, we created a while loop that continuously checks the distance. In this while loop we also added an if condition to check the distance between the user and the obstacle. If it detects an object close to the user, it will automatically inform the person by giving audio and haptic feedback.

- **Code for Person Recognition:**

This is to recognize the person in front of the user (known or unknown). Here in this code, we will import 3 modules: face recognition, cv2 and numpy. We have to create different arrays for known faces and their names. We must write the image file name for the face recognition of that member.

In the next part of code, we will try to match the face in camera video with the array of known faces.

Once all the code is done, we'll make the required connections with all the components. Then we'll run both codes and our eyeglasses will start working.

4. PROGRAMMING ENVIRONMENT & TOOLS:

- Raspberry PI
- Camera
- Ultrasonic sensor
- USB Cable
- Wires
- Glass
- Battery
- OpenCV
- Python3
- Numpy

5. REFERENCES:

[1] https://www.who.int/news-room/fact-sheets/detail/blindness-and-visualimpairment#:~:text=Globally%2C%20at%20least%202.2%20billion,near%20or%20distance%20vision%20impairment.

[2] Ultrasonic Smart Spectacles for Visually Impaired and Blind People **Authors:**

L. Shashitha

(Department of Electronics and Communication Engineering, Institute of Aeronatical Engineering College Dundigal, Hyderabad, Telangana)

P. Ashok Babu

(Department of Electronics and Communication Engineering, Institute of Aeronatical Engineering College Dundigal, Hyderabad, Telangana)

Published in: 2021 3rd International Conference on Advances in Computing, Communication Control and Networking (ICAC3N)

Date of Conference: 17-18 December 2021
Date Added to IEEE *Xplore*: 09 March 2022
Electronic ISBN:978-1-6654-3811-7

Print on Demand(PoD) ISBN:978-1-6654-3812-4

INSPEC Accession Number: 21666268

DOI: 10.1109/ICAC3N53548.2021.9725747

Publisher: IEEE

Conference Location: Greater Noida, India